

CHAPTER 3

DEEP LEARNING BASED VIRTUAL SENSING FOR AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE

This chapter presents an approach that uses VS and DL methods for real-time WQM. The fusion of the cost-effectiveness of VS with the accuracy of DL results in a cost-efficient yet precise solution for real-time WQM. This methodology is rigorously tested using two separate datasets, ADAK and MAC, demonstrating its efficacy and promising capabilities. In this approach, the intent is to replace high-priced DO sensor with virtual reducing the overall cost of the WQM system. Section 3.1 of this chapter serves as an introductory section, outlining the key contributions of this work. Section 3.2 of this chapter explains the proposed DL enhanced VS methodology for WQM. Section 3.3 explains in details the experimental setup for testing the proposed methods. Section 3.4 presents the final results and comparisons of the VS and DLEVS. The chapter is summarized in Section 3.5.

3.1 INTRODUCTION

Effective management of aquaculture's impact on the environment requires a comprehensive understanding of the ecosystem. WQM provides a holistic view of the ecosystem and is essential for managing aquaculture's impact. Traditional manual inspections may not provide all the required information, and continuous monitoring using sensors is necessary to obtain extensive data that includes all factors affecting the farm [127]. Continuous monitoring provides extensive data that farmers cannot assess manually, enabling the control of the system to reduce the impact of aquaculture on the environment while improving product quality and production quantity.

WQM optimizes resource utilization, enhances sustainability, and reduces wastage, thereby increasing profit margins for farmers. The salinity, pH, DO, and temperature are crucial WQ parameters in aquaculture, and their variation can cause production problems [4]. Fluctuations in any of these parameters can result in significant production losses [128]. Additionally, water contamination due to urbanization and industrialization poses further challenges. DO affects the growth

rate and health of aquatic animals, feed utilization, and can lead to stress and disease when temperature changes abruptly or is chronically near the animals' tolerance levels [129–133].

The traditional means of WQM involve collecting water specimens and sending them to a lab, which can consume time and delay decision-making, impacting productivity [7]. Small-scale farms account for the majority of aquaculture production and can benefit significantly from becoming sustainable and efficient. However, available WQM systems are often not affordable to small-scale farms that contribute significantly to overall production. While some systems for WQM in aquaculture exist, many use probes that are costly [8, 9]. A few works have focused on reducing the cost of WQM systems, including reducing sensor costs [10] and conducting extensive studies on sensors to lower overall system costs [11, 12]. After a careful study, some parameters that can be calculated using other parameters have been identified, enabling the replacement of costly sensors. This system uses solubility equations to calculate DO in water and is modular, portable, and low-cost. It shares information with farmers promptly, enabling them to manage the farm efficiently. Reducing the cost of WQM systems will encourage more farmers to use them and improve overall aquaculture production and sustainability.

This work presents a VS based accurate low-Cost WQM System for Aquaculture, with the following major contributions:

1. Development of a WQM system with improved functionality and efficiency through the incorporation of solar energy harvesting and an energy-efficient scheduling approach. This WQM system served as the foundation and test bed for subsequent research on development of affordable WQM and accurate WQP systems.
2. The application of a novel DLEVS mechanism for aquaculture WQM is introduced, incorporating the strengths of LSTM DL-RNN and VS techniques.
3. LSTM model is developed to predict and compensate potential inaccuracies in DO estimation, which is a crucial component of the aquaculture industry.
4. The performance and efficiency of the DLEVS model is meticulously evaluated with the help of two WQ datasets. It is revealed that this model has the potential to significantly improve the accuracy and reliability of DO estimation in aquaculture environments.

3.2 PROPOSED METHODOLOGY OF AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE

3.2.1 Virtual sensing based water quality monitoring

VS techniques, also called soft sensing, proxy sensing, inferential sensing, or surrogate sensing, are used to provide feasible and economical alternatives to costly or impractical physical measurement instrument. This technique predicts parameters using mathematical models, which are not measured using physical sensors. A VS system uses information available from other measurements and process parameters to calculate an estimate of the quantity of interest. In this work, expensive DO sensor is replaced with solubility equations. In this study, three solubility equations for estimation of DO is investigated.

Weiss Equation [134]: Derived in 1970, Weiss introduced a set of equations to calculate DO as a function of temperature and salinity using the method of least squares to fit data collected on the solubility of nitrogen, oxygen, and argon in water and seawater. The DO at zero salinity, and one atmosphere is given in Eq. (3.1). The Weiss salinity factor is given in Eq. (3.2). The equation for calculating DO is given in Eq. (3.3).

$$DO_o = 1.42905 \exp \left[-173.4292 + 249.6339 \left(\frac{100}{T} \right) + 143.3483 \left(\ln \frac{T}{100} \right) - 21.8492 \left(\frac{T}{100} \right) \right] \quad (3.1)$$

$$F_S = \exp \left[S \times \left(-0.033096 + 0.014259 \left(\frac{T}{100} \right) - 0.0017 \left(\frac{T}{100} \right)^2 \right) \right] \quad (3.2)$$

$$DO = DO_o \times F_S \quad (3.3)$$

Benson and Krause Equation [135], [136]: Benson and Krause derived equations based on thermodynamic principles to compute the solubility of oxygen in the water. Their work resulted in equations that provide oxygen solubility as a function of temperature in freshwater and adjustments based on barometric pressure and salinity. Eq. (3.4) is used to compute the DO in mg/L at zero salinity. The Eq. (3.5) gives the salinity factor which is used to estimate the DO when salinity is more than zero. The equation for calculating DO is given in Eq. (3.6).

$$DO_o = \exp \left[-139.34411 + \frac{1.575701 \times 10^5}{T} - \frac{6.642308 \times 10^7}{T^2} + \frac{1.243800 \times 10^{10}}{T^3} - \frac{8.621949 \times 10^{11}}{T^4} \right] \quad (3.4)$$

$$F_S = \exp \left[-S \times \left(0.017674 - \frac{10.754}{T} + \frac{2140.7}{T^2} \right) \right] \quad (3.5)$$

$$DO = DO_o \times F_S \quad (3.6)$$

Garcia and Gordon Equations [137]: Garcia and Gordon re-examined and refitted the data used in many previous works to a higher order polynomial to estimate the DO when salinity is more than zero. The Eq. (3.7) gives scaled temperature which is used in Eq. (3.8) and Eq. (3.9). The Eq. (3.9) gives the salinity factor which is used to

estimate the DO when salinity is more than zero. The equation for calculating DO is given in Eq. (3.10).

$$T_s = \ln \left[\frac{298.15 - t}{273.15 + t} \right] \quad (3.7)$$

$$DO_o = 1.42905 \exp \left[2.00907 + 3.22014T_s + 4.05010T_s^2 + 4.94457T_s^3 - 0.256847T_s^4 + 3.88767T_s^5 \right] \quad (3.8)$$

$$F_S = \exp \left[(-0.00624523 - 0.00737614T_s - 0.0103410T_s^2 - 0.00817083T_s^3) S - 4.88682 \times 10^{-7} S^2 \right] \quad (3.9)$$

$$DO = DO_o \times F_S \quad (3.10)$$

3.2.2 Deep learning enhanced virtual sensing for water quality monitoring

This section explains the use of DL to reduce the error in the VS based estimation of the DO using the solubility equation. DL, specifically an LSTM model, is utilized to improve the accuracy of the VS-based estimation of DO. This is achieved by predicting the error in VS using the LSTM model, then subtracting this error from the DO value estimated using VS, thus enhancing the accuracy of DO estimation. An error prediction model is initially developed, which predicts the error using WQ time sequence data collected via the WQM system. The predicted error is then used to reduce the error in VS and improve the accuracy of VS. The error at any point in time is calculated as the difference between the actual value and the VS values, as shown in Eq. (3.11).

$$DO_{er}(t) = DO_{vs}(t) - DO(t) \quad (3.11)$$

Eq. (3.11) $DO_{vs}(t)$ is the estimated DO using solubility equations with temperature and salinity data at time t . $DO(t)$ is the original value of DO in the data which is measured using the DO sensor at time t . $DO_{er}(t)$ is the estimated error at time t . The model uses the error dataset to train the LSTM model and predict the error. Figure 3.1 shows how LSTM model is used predict the error in VS of DO. Here the error time sequence WQ dataset is used for training the LSTM model. For making the error prediction, the error in VS is estimated using the datasets and this error data is used to train the LSTM model. After training the LSTM model using the time sequence error data, the model predicts the error which is used to reduce the error in VS. The error predicted by the LSTM model is used for reducing the error in VS. Here the error in VS is minimized to make the system more accurate and reliable. LSTM model predict the error in the system, which can be feedback to the system to reduce the error during estimation of DO. Figure 3.2 shows the block diagram of the error reduction in VS using LSTM model. Here T , S is the temperature and salinity values which are used for VS. T is the temperature dataset $T(0)$ to $T(t-1)$, S is the salinity dataset $S(0)$ to $S(t-1)$, DO is the DO dataset $DO(0)$ to $DO(t-1)$, DO_{er} is the error dataset $DO_{er}(0)$ to $DO_{er}(t-1)$, DO_{er} is given by the Eq. (3.11). The LSTM model predicts

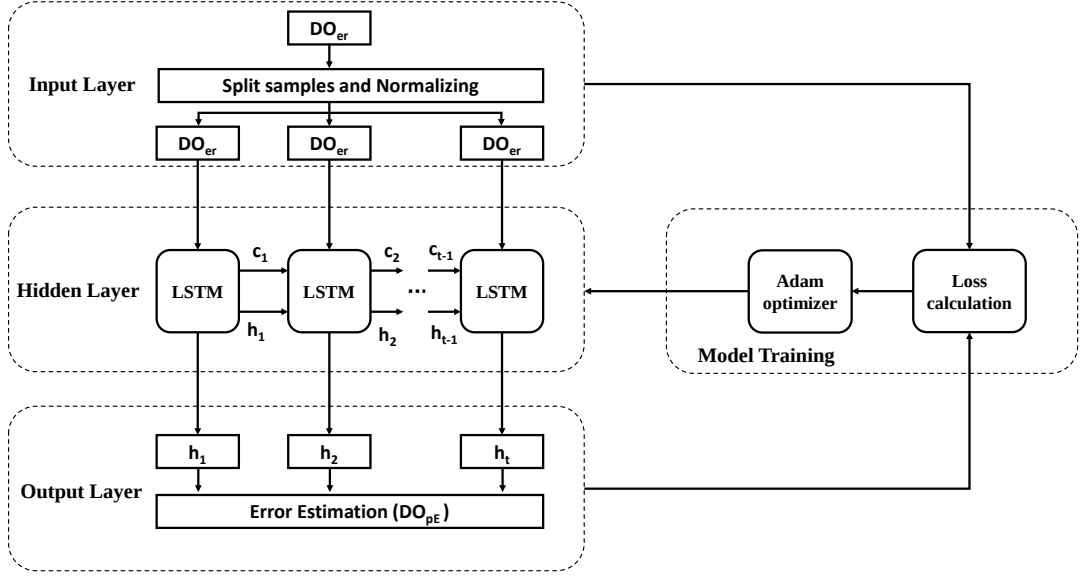


Figure 3.1: Error prediction system based on LSTM model

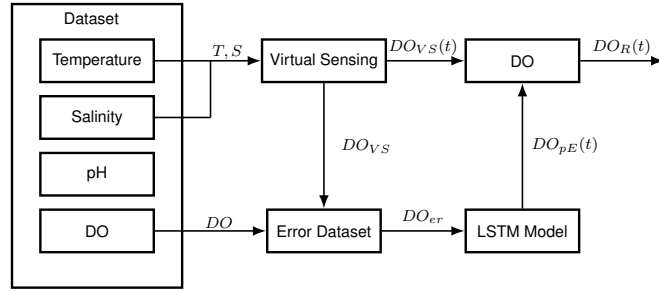


Figure 3.2: Block diagram of DLEVS

the error in VS and is denoted as DO_{pE} . Predicted error DO_{pE} improve the accuracy of the VS by deducting the same from DO_{VS} to give DO_R as in Eq. (3.12). DO_R is the DO which is computed finally after using LSTM to reduce the error of the VS.

$$DO_R(t) = DO_{vs}(t) - DO_{pE}(t) \quad (3.12)$$

3.3 EXPERIMENTAL SETUP FOR TESTING THE PROPOSED METHODS

This section presents the characteristics of the proposed IoT-based WQM system for aquaculture. The entire system is a WSN-based system where Node S collects data from a distributed network of sensor nodes deployed on a wide field. Node S is a solar-powered sensing unit that collects different WQ parameter data and transmits them to a central node, Node C. The proposed virtual sensing is implemented in Node S, a low-cost and low-power device. The design of this system requires the selection of important WQ parameters since using more sensors will make the system costly. The key parameters are temperature, pH, and salinity. Using them, DO can be estimated by

virtual sensing.

A WQM system requires the development and integration of hardware and software components. Figure 3.3 shows the overall architecture of the developed WQM system. The first part of the system is Node S. Node S comprises the sensors responsible for detecting WQ parameters and converting them into electrical signals. Another component of Node S is the Microcontroller, which converts the electrical signals or voltage levels from sensors into data. Following Node S, there is Node C, tasked with coordinating all Node S units. It gathers data from each Node S, stores the collected data, and facilitates information communication. It serves as the central hub of the system, hence its designation as Node C.

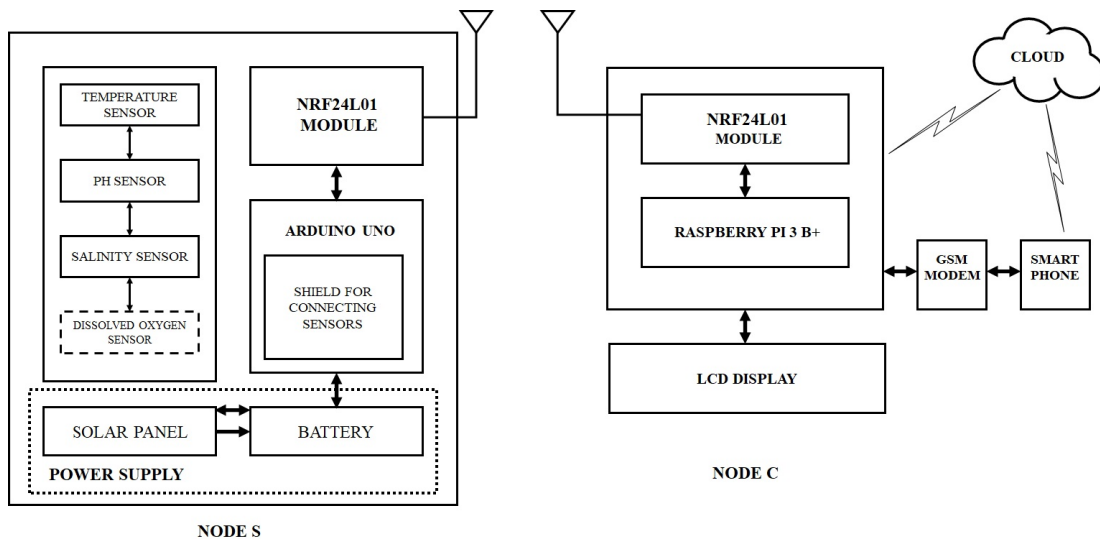


Figure 3.3: WQM system architecture

3.3.1 Hardware

Node S and Node C constitute the two fundamental modules of the hardware. This system can accommodate a variable number of Node S units and one Node C responsible for controlling and gathering data from all the Node S units.

Node S: Node S is comprised of four subsystems: the sensor part, the microcontroller (Arduino Uno), the communication subsystem (NRF24L01), and the power supply. The WQM sensors incorporated in this node encompass temperature, pH, and salinity. To estimate DO, the Weiss equation is employed within the microcontroller. Each sensor can detect specific WQ parameters and convert them into electrical signals. The microcontroller executes programs that translate these electrical signals into corresponding WQ parameters. The microcontroller chosen for this purpose is the ARDUINO UNO REV3, which is built around the ATmega328P microcontroller chip, providing ease of use and compatibility with numerous shields

designed for Arduino Boards. The power supply is equipped with a 20 ampere-hour battery and an 18-watt solar panel to ensure continuous operation of the sensor node throughout the year, eliminating the need for frequent battery recharging or replacement. Table 3.1 lists the sensors employed in Node S.

Table 3.1: Sensors used for measuring the water quality parameters

Sensor	Units	Range	Sensor Type
pH		0 - 14	Gel-filled(Ag/AgCl)
Temperature	°C	-40 - 135	Thermistor
Salinity	ppt	0 - 50	Parallel platinum electrode
DO	mg/L	0 - 20	Luminescent

Node C: Serving as the core of the system, Node C handles the task of aggregating data from Node S units (retrieved via sensors, the Arduino microcontroller, and the NRF24L01 transceiver). It archives data for prolonged periods, analyses data, disseminates information to farmers, and uploads data to the cloud as required. The Raspberry Pi 3 B+ is utilized as the controller within Node C. It is interconnected with the NRF24L01 transceiver to retrieve data from Node S and to manage Node S operations.

3.3.2 Software

Once the hardware of the system is prepared, the configuration and installation of the necessary software, along with the upload of codes for proper operation of Node S, require the utilization of the Arduino IDE. Node C functions as a central unit, gathering all the data. The software aspect encompasses two components: one for Node S and the other for Node C. A dedicated custom shield is employed with the Arduino Uno to capture analogue signals from the sensors. A monitoring program, referred to as a sketch, identifies each probe by comparing its signature voltage range and subsequently converts the analogue electrical signal into meaningful data. The Arduino IDE facilitates the uploading of these sketches onto the Arduino Uno, enabling the sketches to read the sensors and collect data. Multiple sensors can be connected to a single board, enhancing versatility. The Raspberry Pi (RPi) board accumulates data from various Arduino boards. As the central node, the RPi is an optimal platform to implement decision support mechanisms. For instance, it determines critical DO levels that require immediate farmer notification, in addition to generating periodic reports.

The Arduino retrieves information from the sensor node and interacts with farmers via Node C, which serves as a communication gateway for various forms of communication. Each sensor gathers data at scheduled intervals, facilitated by Arduino. The Arduino establishes direct communication with the sensor through its analogue pins,

while communication with Node C is achieved via the NRF24L01 transceiver. Using data from temperature and salinity sensors, the Arduino employs the Weiss Equation to estimate the DO content in water. It processes the electrical signal received from the sensors to generate the necessary data. After acquiring data from Node S at specific intervals, the Arduino relays the information to farmers through Node C. The timing of these alerts is predefined within Node C. Acting as a bridge between Node C and sensors, the Arduino operates as a central communication hub. The architecture follows a modular design, facilitating system expansion. This adaptability makes the system suitable for deployment in both small and large farms.

3.4 RESULTS AND DISCUSSIONS

3.4.1 ADAK water quality dataset



Figure 3.4: DO estimations (a) using VS and (b) using DLEVS for the ADAK water quality dataset

Figure 3.4a presents a comparison of DO values obtained from the proposed VS model with those gathered from the actual sensor for the ADAK dataset. The comparison demonstrates a close alignment between the predicted and actual values. Furthermore, Fig. 3.4b showcases how integrating the VS model with DL (resulting in a DLEVS system) can further enhance the predictions, bringing them into even closer proximity with the actual values. Table 3.2 offers a comprehensive comparison between the VS and DLEVS methods for various equations. It shows that the DLEVS method significantly enhances the accuracy of predictions across all metrics, outperforming the standalone VS method.

Table 3.2: DO estimation using VS and DLEVS for the ADAK water quality dataset

Equation	VS				DLEVS			
	MAE	MSE	RMSE	MAPE	MAE	MSE	RMSE	MAPE
Weiss	0.3038	0.1445	0.3802	0.0417	0.1163	0.0235	0.1534	0.0158
Benson & Krause	0.3050	0.1542	0.3927	0.0421	0.1157	0.0244	0.1561	0.0156
Garcia & Gordon	0.3071	0.1564	0.3955	0.0425	0.1143	0.0230	0.1516	0.0155

3.4.2 MAC water quality dataset

Figure 3.5, show the comparison of the DO values from the proposed VS and DLEVS methods with the actual sensor's readings for the MAC dataset. Specifically, Fig. 3.5a displays the comparison for the VS method, whereas Fig. 3.5b presents the comparison for the DLEVS method. From these figures, it can be seen that the proposed methods align well with the actual DO values. Moreover, the DLEVS method demonstrates a closer agreement with the actual values than the VS method does, as evidenced by the smaller errors presented in Table 3.3.

The results for both datasets indicate that the proposed method of combining VS with DL models (DLEVS) outperforms the traditional VS for estimating DO levels in water. The DLEVS method consistently achieved lower MAE, MSE, RMSE, and MAPE values across all three DO equations for both the ADAK and MAC datasets. The results suggest that incorporating DL models into the virtual sensor framework can lead to more accurate and reliable estimates of WQ parameters.

Table 3.3: DO estimation using VS and DLEVS for the MAC water quality dataset

Equation	VS				DLEVS			
	MAE	MSE	RMSE	MAPE	MAE	MSE	RMSE	MAPE
Weiss	0.6316	0.5633	0.7505	0.1002	0.0567	0.0072	0.0848	0.0089
Benson & Krause	0.6531	0.5936	0.7705	0.1038	0.0590	0.0074	0.0862	0.0092
Garcia & Gordon	0.6474	0.5855	0.7652	0.1029	0.0587	0.0076	0.0870	0.0091

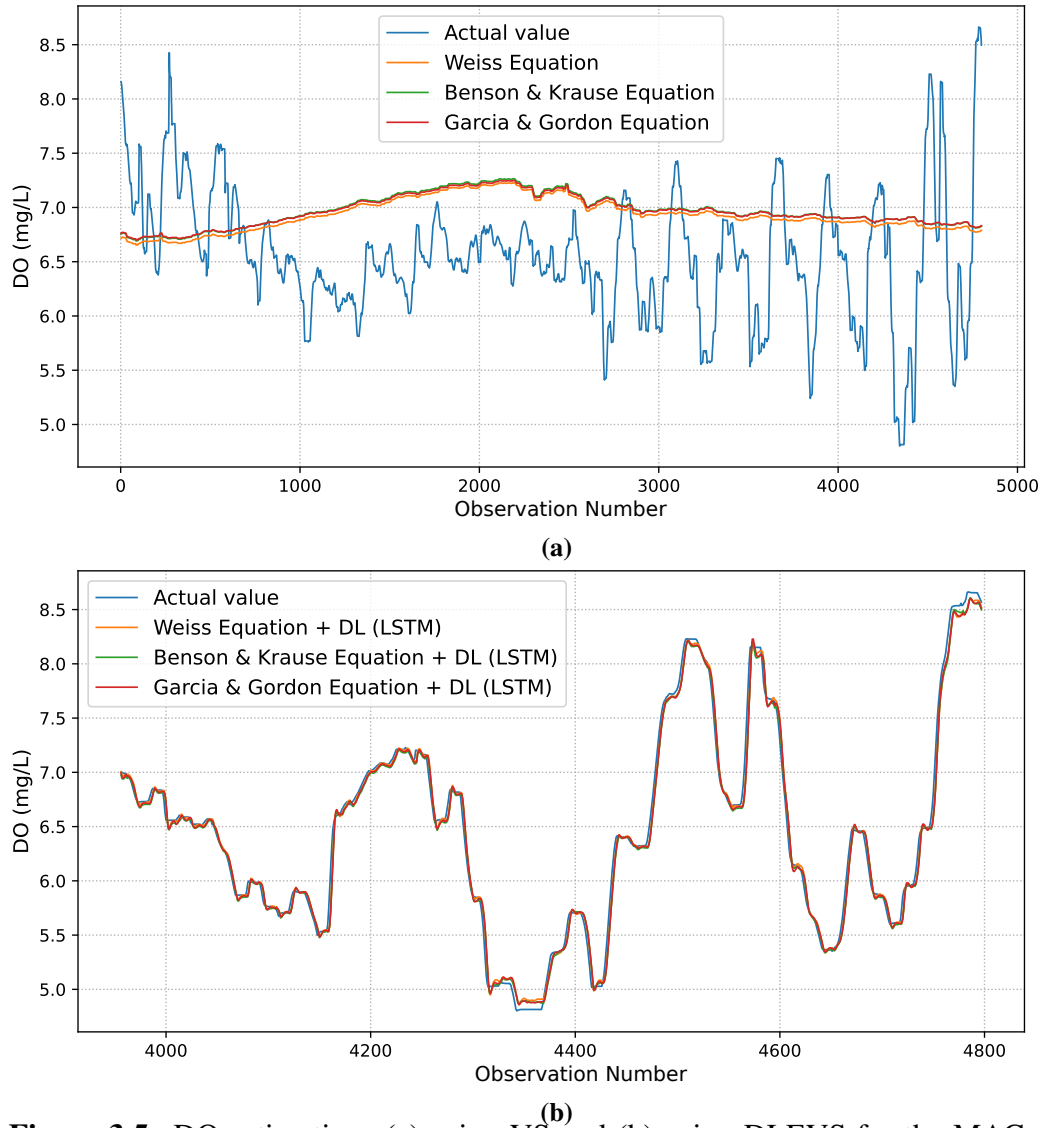


Figure 3.5: DO estimations (a) using VS and (b) using DLEVS for the MAC water quality dataset

3.5 SUMMARY

In this research work, a low-cost WQM system is proposed for aquaculture, which utilizes a virtual sensing approach based on the LSTM DL-RNN. The virtual sensor estimates the DO value using various solubility equations, and the LSTM DL-RNN is used to reduce the error in DO estimation, thus improving the accuracy of the virtual DO sensor. The results demonstrate that the proposed LSTM-based DO virtual sensor can accurately estimate DO levels, providing a cost-effective and reliable solution for monitoring WQ parameters in aquaculture. This research work paves the way for the development of low-cost and efficient WQM systems for aquaculture, which are very important for ensuring the sustainability and productivity of Aquaculture industry.